# Final Presentation — Speaker Notes

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**Structure:** 26 main slides + 4 backup. Framing throughout is paper vs our implementation. No internal-iteration history in what the audience hears.

### Slide 1. Title

- Final presentation of our paper reproduction project — Shi, Lee, Yang (IEEE CASE 2024).
- Presentation structure: recap, reproduction scope, implementation requirements, reproduction results, extensions, and summary.

### Slide 2. Recap: Simulating Emergency Evacuation

- Brief recap from our 1st presentation. The paper studies how thousands of pedestrians evacuate when a hazard affects a community.
- The slide only lists the three main reasons this is hard: independent decisions, local information, and emergent panic/congestion. Say the fuller version verbally: each person chooses a destination and route, those choices depend on nearby roads and hazard exposure, and congestion is not imposed globally but emerges from many individual movements.
- Therefore a traditional process-flow model is too aggregated. The paper uses agent-based simulation so each pedestrian has its own state, position, speed, destination, and decision logic.

### Slide 3. Recap: Three-Component Model

- Three coupled components from the paper: community network, human agents, hazard agents.
- Two algorithms tie them together — Algorithm 1 (human–network) and Algorithm 2 (human–hazard). Both run every step.

### Slide 4. Recap: Agent States & Algorithms 1–2

- Six states per human, with Algorithm 2 handling hazard exposure and Algorithm 1 handling routing. The slide is intentionally compressed; give the operational details verbally.
- Algorithm 2 runs every step: if an agent is inside a hazard for the first time, it becomes impacted and reroutes to a shelter; then casualty risk and panic are checked. Casualty probability is higher near the hazard center.
- Algorithm 1 also runs every step: agents on edges check congestion and may queue; agents at nodes reroute, continue toward a destination, or choose a random route if panic has taken over.

- Two points worth emphasizing: once impacted, destination flips to the nearest shelter; panic is irreversible.
- We take both algorithms directly from the paper — no changes to the decision logic.

**Slide 5. Reproduction and extension scope**

- Three reproduction targets from the paper: Phase 1 (five community networks), Phase 2a ($RI$ vs $P_{\max} \times H_{\max}$), Phase 2b ($RS/RC/RL$ vs panic).
- If needed, expand the abbreviations verbally: $RI$ is impacted rate, $RS$ is survival rate, $RC$ is casualty rate, and $RL$ is leftover rate, meaning agents still stuck in hazard at the end.
- Three extensions beyond the paper's range: wider $P_{\max}$, wider $H_{\max}$, finer panic grid.
- The notation box at top defines every symbol used later.

**Slide 6. Implementation specifications and reproducibility gaps**

- The paper specifies the implementation environment, hardware, and measured runtime, but it does not fully specify the computational implementation. Algorithms 1 and 2 define decision logic, not the path-computation mechanism.
- The main unspecified details are shortest-path implementation, per-step versus cached path policy, parallel execution, RNG design, and reproducibility seeding. These choices materially affect whether a multi-seed reproduction is feasible.
- Key takeaway: at the paper's 2.5 h/run baseline, our 390-run core grid would require about 41 days of sequential compute. The optimized pipeline completes that grid in 3.4 hours, making the subsequent reproduction and extension analyses practical.

**Slide 7. Computational bottleneck: repeated path computation**

- Algorithm 1 path recomputation dominates runtime: 2,000 agents across 121 steps is roughly 242K shortest-path queries per run.
- Optimization: many agents share destinations, so instead of asking the graph for a new path for every agent, we group by destination, build one predecessor tree for each destination, and reconstruct each agent's path from that tree.
- The lower summary box reports the measured runtime change: the routing subsystem drops from 165.9 seconds to about 20.0 seconds including tree construction, and the full run drops from 168 seconds to 22.5 seconds.

**Slide 8. Experimental design for scalable reproduction**

- Three implementation decisions support scalable reproduction: recompute paths every step, execute the grid with 8 workers, and vary agent seeds while keeping the hazard seed fixed.
- These are implementation choices, not new behavioral rules. They preserve the original model logic while making multi-seed reproduction possible.

**Slide 9. Experimental scale enabled by the implementation**

- The optimized implementation supports four analysis components: 10 seeds per configuration, a 390-run core grid, a 510-run Pmax sweep, and three visual comparison animations.
- The gray caption reports the measured speedup: 7.5x single-process and about 60x with 8 workers.

**Slide 10. Baseline PSU-UP simulation on the reproduced network**

- Animation on the paper's baseline ($P_{\max} = 2{,}000$, $H_{\max} = 5$, panic 10%).
- In the animation, hazards appear, impacted agents reroute to shelters, some become casualties, and the remaining agents arrive. Final $RI$ is about 36%, matching the Phase 2a chart on slide 13.

**Slide 11. Phase 1: Network validation**

- Building counts match the paper's Table V within 0–2% across all five communities.
- Edge counts: PSU-UP, UVA-C, VT-B line up. RA-PA and KOP-PA are anomalous — paper's $E/N$ ratios of 6.5–6.8 are unreproducible with any OSMnx configuration we tried. Treated as a paper data anomaly; details in backup slide 29.

**Slide 12. Phase 1: PSU-UP figures**

- Top: Fig. 3 reproduction — the PSU-UP walk network, with 969 building nodes and the supplemented shelter set fixed before later experiments.
- Bottom: Fig. 4 reproduction — pedestrian flow snapshots at five times during a hazard scenario. The key visual check is that impacted agents appear after hazards emerge, then most agents move into survival or arrival states by the end.

**Slide 13. Phase 2a: $RI$ vs $P_{\max} \times H_{\max}$**

- Fig. 5 reproduction. Our qualitative trend matches: $RI$ increases with $H_{\max}$ and is nearly independent of $P_{\max}$ in the paper range.
- Critical confirmation: $RI$ is independent of $P_{\max}$ from 2K to 8K (delta under 0.2%p). This is the paper's core finding, and we reproduced it.
- Quantitative match is partial, not exact: mean absolute error is about 5 percentage points and the largest gap is about 9 percentage points.

**Slide 14. Phase 2b: $RS/RC/RL$ vs panic**

- Fig. 6 reproduction using the paper's five panic points. All three directions match: higher panic lowers survival and raises casualty and leftover.
- Quantitative levels are only partially matched. The largest visible gap is high-panic casualty: our implementation levels off near 6%, while the paper reaches about 13%.
- Therefore the academically precise interpretation is qualitative reproduction with remaining model gaps, rather than exact numerical reproduction.

**Slide 15. Calibration choices and remaining gaps**

- The slide is intentionally simpler than the notes: the three main unpublished inputs are shelter locations, casualty formula, and hazard seed. We calibrated each once, then held it fixed for the rest of the deck.
- Shelter locations matter most for survival. The table shows why we use about 600 shelters: it gives 64.3% survival at panic = 90%, closest to the paper's 64.6%.
- The remaining gaps should be stated explicitly: $RI$ is a little high overall, and high-panic casualty remains low because our casualty formula is conservative relative to the paper's implied behavior.

**Slide 16. Extension study design beyond the paper range**

- The optimized pipeline supports three controlled extensions beyond the paper range.
- The paper tested $P_{\max}$ up to 8K, $H_{\max}$ up to 15, and panic at five points. We extend each axis in the next three slides.

**Slide 17. Extension 1: $P_{\max}$ threshold**

- Extended $P_{\max}$ from 2K to 97K across three $H_{\max}$ levels.
- Independence holds up to 20K, and a measurable deviation appears at 50K by +8.5%p.
- Mechanism hypothesis: at 50K, building-to-network connectors and nearby walk edges become bottlenecks, so more agents queue outdoors and remain exposed longer.

**Slide 18. Extension 2: $H_{\max}$ saturation**

- Extended $H_{\max}$ from 5 to 30 at paper-baseline $P_{\max}$.
- $RI$ approaches an empirical plateau near 86% by $H_{\max} = 30$.
- Interpretation: under this hazard-generation process, the remaining 14% appears to be low-exposure or peripheral origins, not a universal ceiling.

**Slide 19. Extension 3: Finer panic grid**

- 9 panic points vs the paper's 5.
- Curves are smooth and monotonic at 10% resolution; no abrupt transition is visible in this calibrated scenario.
- Model implication: marginal panic reduction has the largest effect at low baseline panic.

**Slide 20. Visual comparison 1: $P_{\max}$ 2K vs 50K**

- Same hazard scenario, different $P_{\max}$. Emphasize outdoor queuing in the high-$P_{\max}$ panel.

**Slide 21. Visual comparison 2: panic 10% vs 90%**

- Same hazard, different panic rate. High-panic agents select random routes more often instead of moving directly toward shelters.

### Slide 22. Visual comparison 3: $H_{\max}$ 5 vs 30

- Hazard counts differ by design — say this explicitly. Left has 5 hazards, right has 30.
- The right panel shows substantially broader hazard coverage, so agents are exposed repeatedly.

### Slide 23. Summary

- Summary structure is now three columns: reproduction, engineering, and extensions.
- The conclusion: the paper's qualitative behavior is reproduced, and the scalable pipeline enables controlled follow-up experiments without changing the behavioral state logic.
- The three-number dashboard keeps only communities, total simulation runs, and overall speedup.

### Slide 24. Scope and limitations

- Analysis scope: faithful paper reproduction, an implementation improvement, and three controlled extensions.
- Out-of-scope extensions: agent-model changes, casualty-formula rewrite, and multi-community extension of Phase 2. These remain future-work directions because they change the model layer rather than only the experimental design.

### Slide 25. Team contributions

- This slide satisfies the project guide requirement to delineate each member's contribution in the report.
- Keep the delivery brief: individual technical roles first, then shared work across environment setup, debugging, result checking, presentation preparation, README writing, final code review, and final report approval.

### Slide 26. References & acknowledgements

- Acknowledge the primary paper, the 2021 companion paper we used to disambiguate preprocessing, OSMnx, and SciPy.
- Code and data on GitHub. Thank the audience; invite questions.

### Slide 27. Backup: Future work

- If asked about future work: the paper's own suggestions — response agents, social-force, MDP panic model — plus our own observation that a panic-dependent casualty formula could close the $RC$ ceiling gap.

### Slide 28. Backup: $P_{\max}$ threshold generalization

- If asked whether the high-$P_{\max}$ threshold is specific to PSU-UP: multi-community $P_{\max}$ data is in this table. Pattern is consistent across communities with different building counts.

**Slide 29. Backup: Edge count anomaly**

- If asked about the RA-PA and KOP-PA mismatch: this backup has the diagnostic table. Paper reports $E/N$ of 6.5–6.8, which is implausibly high for pedestrian walk networks and unreproducible with any standard OSMnx configuration.

**Slide 30. Backup: Static snapshots**

- If the venue's PDF viewer does not play the embedded animations: these still frames are the fallback.